Answer Key - Chemistry - Graduate Level Assessment 1 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) 3 atm; B) 9 atm; C) 5 atm; D) 6 atm; E) 10 atm
   Note: Correct option: C - 5 atm

2. **Answer:** E
   Options: A) 7.50 10^{-3}%; B) 2.50 10^{-3}%; C) 0.75 10^{-3}%; D) 2.00 10^{-3}%; E) 1.25 10^{-3}%
   Note: Correct option: E - 1.25 10^{-3}%

3. **Answer:** A
   Options: A) Add 67.7 mL of 12.0 M acid to the flask, and then dilute it with 33.3 mL of water.; B) Add 33.3 mL of water to the flask, and then add 66.7 mL of 12.0 M acid.; C) Add 40 mL of water to the flask, and then add 60 mL of 12.0 M acid.; D) Add 60 mL of 12.0 M acid to the flask, and then dilute it with 40 mL of water.; E) Add 50 mL of 12.0 M acid to the flask, and then dilute it with 50 mL of water.
   Note: Correct option: A - Add 67.7 mL of 12.0 M acid to the flask, and then dilute it with 33.3 mL of water.

4. **Answer:** D
   Options: A) pH is 10, pOH is 4; B) pH is 4, pOH is 10; C) pH is 3, pOH is 3; D) pH is 4, pOH is 4; E) pH is 1, pOH is 1
   Note: Correct option: D - pH is 4, pOH is 4

5. **Answer:** A
   Options: A) - 13.61 eV; B) 1.6022 × $10^{-}19C; C) -2.1802 \times 10^{-}18J; D) 9.1095 \times 10^{-}31kg$
   Note:$Correctoption: A - -13.61eV$

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** C_6H_7Br. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 2.10 gallons. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key